THE CALABI–YAU-TO-CHIRAL FUNCTOR Φ AND THE UNIVERSAL TRACE IDENTITY

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Abstract. The Calabi–Yau-to-chiral functor $\Phi\colon \mathrm{CY}_d \to \text{Chiral-Alg}$ assigns to a compact Calabi–Yau d-fold X a chiral factorisation algebra $\mathcal{A}_X$ on an auxiliary curve. Its principal invariant is $\kappa_{\mathrm{ch}}(\mathcal{A}_X) = \sum_q (-1)^q h^{0,q}(X)$ (Hodge supertrace, proved with separate algebraic and geometric models). The output E_n-scope stratifies by dimension: $\Phi(X) \in E_2$-chiral for $d \le 2$ and $\Phi(X) \in E_1$-chiral for $d \ge 3$. The universal trace identity unifying Vol I $K = -c_{\mathrm{ghost}}(\mathrm{BRST})$ with $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ connects Vol I's universal conductor with Vol III's Borcherds weight.

Scope and status

Primary theorems:

- Φ preserves Koszul duality on the Koszul locus: .
- $\kappa_{\mathrm{ch}} = \chi_{\mathrm{Hodge}}$ (Hodge supertrace): (`thm:kappa-hodge-supertrace-identification`).
- E_n-output scope $n = 2$ for $d \le 2$, $n = 1$ for $d \ge 3$: (FM43 scope guard).
- Universal trace identity $K_{\mathrm{univ}} = -c_{\mathrm{ghost}}$: (Vol I `chap:universal-conductor`).

Chapter inputs

The standalone records the following chapter sources:

```
\input{../chapters/theory/cy_to_chiral.tex}
\input{../chapters/theory/quantum_chiral_algebras.tex}
\input{../chapters/theory/en_factorization.tex}
\input{../chapters/examples/cy_d_kappa_stratification.tex}
```

Cross-volume dependencies

- Vol I `chap:universal-conductor` (universal trace identity $K_{\mathrm{univ}} = -c_{\mathrm{ghost}}$).
- Vol I `thm:A-infinity-2` (properad bar-cobar; Vol III Φ is a corollary of Francis–Gaitsgory (`cor:chiral-KK-formal-smoothness`).
- Vol II `thm:chiral-higher-deligne` (target category for $\Phi(X)$'s derived centre at $d = 3$).

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